Melco Training Overview

Yellowbrick Data

Melco Training

Connect. Query. Report. Repeat.

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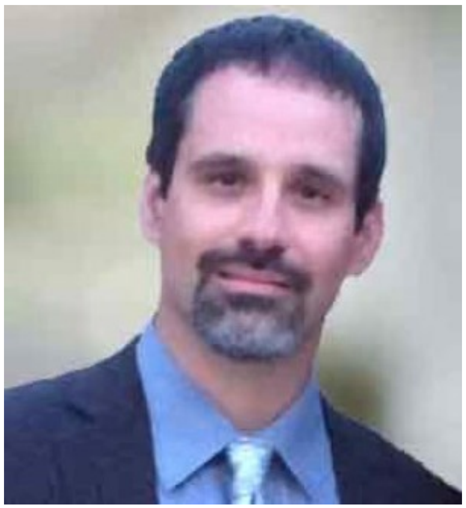
About Me



- Michael Suarez
- Sr. Director, Solutions Engineering
- 4 years with Yellowbrick
- 25+ Years in Analytics and Data Warehousing
- Live in Ann Arbor, Michigan

Melco Training Objectives

The course aims to improve the productivity and independence of Melco's analysts and business users on Yellowbrick — enabling them to connect, query, manage, and report on their own data.

- **Get Oriented:** platform architecture, the Manager GUI, and how databases, schemas, and users fit together
- **Connect and Query:** link client tools, write SQL to answer business questions, export results for reporting
- **Operate and Apply**: data pipelines, stored procedures, SAS migration, and self-service data management
- **Administer and Integrate**: security, monitoring, performance tuning

Day 1 — BU | Foundations & Getting Started

Time	Session
10:00–11:00	Welcome & training overview; What is Yellowbrick (analytic vs OLTP, MPP, shared-nothing, column store, cloud / on-prem)
11:00–11:10	*Morning break*
11:10–12:00	Module 2 – Architecture: MPP compute nodes, data distribution & skew, join co-location, block pruning; scalability & storage layers
12:00–1:00	*Lunch*
1:00–2:00	Demo: Getting started: navigating the Yellowbrick Manager GUI (instance & cluster management)
2:00–2:10	*Afternoon break*
2:10–3:00	Module 3 – Databases, Schemas & Tables: object hierarchy, data types. Intro to User Roles.
3:00–3:10	*Afternoon break*
3:10–4:00	Module 3 Exercises: explore the object hierarchy; create schemas & tables with distribution and ordering keys; query samples.retail data

Day 2 — BU | Connecting, SQL & JSON

Time	Session
10:00–11:00	Module 4 – Connecting to Yellowbrick: ybsql, DBeaver, ODBC/JDBC; connection security & best practices
11:00–11:10	*Morning break*
11:10–12:00	Module 5 – Efficient SQL: SELECT, WHERE, ORDER BY, aggregation (GROUP BY, SUM, COUNT), JOINs, CTEs & Window Functions
12:00–1:00	*Lunch*
1:00–2:00	Module 5 Exercises: Writing Efficient SQL exercises
2:00–2:10	*Afternoon break*
2:10–3:00	Module 6 – Working with JSON: JSONB storage, extracting values, filtering & flattening nested structures with LATERAL
3:00–3:10	*Afternoon break*
3:10–4:00	Module 6 Exercises: Working with JSON in Yellowrbrick

Yellowbrick

Day 3 — BU | Data Pipeline, Logic & Performance

Time	Session
10:00–11:00	Module 7 – Data Ingestion: SQL INSERT, COPY, LOAD TABLE (S3/Azure/GCS), ybload; bulk vs trickle strategies & error handling
	11:00–11:10 Morning break
11:10–12:00	Module 8 – Reusable Logic: Views, Stored Procedures & SAS-to-Yellowbrick migration patterns; Python for analytics
	12:00–1:00 Lunch
1:00–2:00	Module 8 Exercises: Stored Procedures and Reusable Business Logic
	2:00–2:10 Afternoon break
2:10–3:00	Module 9 – Reading Query Plans: EXPLAIN & EXPLAIN ANALYZE; plan nodes, data movement, optimization red flags & rewrites
	3:00–3:10 Afternoon break
3:10–4:00	Module 10 – Training Summary: 3-day recap of all 9 modules; key concepts review & open Q&A

Melco Training - Feedback

- This content has been produced specifically for this audience
- Represents a big investment of time for students and we want to make the most of it
- Please provide feedback early and often so we can make the most of this experience.

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Links and Contact Information

CONTENT	LINK	COMMENTS
Product Documentation:	docs.yellowbrick.com	Searchable online Product Documentation
Yellowbrick Support Center:	support.yellowbrick.com	Support services, ticketing, knowledge base, and downloads.
Corporate Web Site:	yellowbrick.com	Yellowbrick Data public web site

Yellowbrick

What is Yellowbrick?

A Columnar MPP Analytic Data Warehouse in the Cloud and On-Premises

Yellowbrick is a Data Warehouse

Data Warehouse | Analytic Database | MPP | Shared Nothing | Column Store | On-Prem & Cloud

A repository for loading, transforming, and querying large volumes of data in big, set-based operations.

- Consolidates data from multiple source systems into one place.
- Retains long historical depth, not just current-state data.
- Uses schemas designed for analysis (e.g., star/snowflake) rather than transactional normalization.
- Serves as a single source of truth for reporting and analytics across the organization.

Yellowbrick is an Analytic Database

Data Warehouse | **Analytic Database** | MPP | Shared Nothing | Column Store | On-Prem & Cloud

Built for analytical queries against very large datasets.

- Not an OLTP system: no fine-grained transaction control, triggers, or rapid-fire small operations.
- Designed primarily for read-and-append workloads; updates and deletes are supported but not optimized—expect poor performance.
- Requires set-based design—ELT and stored procedures, not row-by-row logic.
- Performs poorly when OLTP habits are ported over, such as:
  - Loops and parallelized statements that touch only a few rows.
  - Continuously deleting the oldest records from fact tables.

Analytic (OLAP) vs Transaction (OLTP) Databases

	OLTP Database	OLAP Database
Optimized For	large volume high concurrency of short, simple read and write transactions, with small amounts of data	Optimized for long-running, complex, read-heavy queries and bulk loads involving large datasets
Txn Handling	Optimized for many short, concurrent INSERT/UPDATE/DELETE transactions with strict ACID guarantees	Simplistic in comparison to OLTP dbs due to intended use but still ACID compliant.
Data Storage	Typically row-oriented storage optimized for fast single-row access and updates	Often column-oriented storage optimized for compression and large-scale scans
Constraints	Enforces PKEY, FKEY, UNIQUE, and referential integrity constraints for transactional consistency	Most Constraints are informational-only to maximize bulk load and query performance

Yellowbrick is an MPP Database

Data Warehouse | Analytic Database | **MPP** | Shared Nothing | Column Store | On-Prem & Cloud

Massively Parallel Processing (MPP) databases use an architecture of multiple nodes working together to process queries and data.

What this means:

- Data and work are distributed across the workers; each worker does its share of work on the data it has
- Some steps can only be done on a single worker though. E.g. the final step of an aggregation or sort
- The performance of a statement is determined by the slowest worker.
- The system is effectively out of storage when any worker is out of storage.

Yellowbrick uses a Shared Nothing Architecture

Data Warehouse | Analytic Database | MPP | **Shared Nothing** | Column Store | On-Prem & Cloud

Each Compute Node has autonomy over its subset of data.

What this means:

- This is the real magic of an MPP system that enables it to parallelize work in an efficient way.
- YB has an "exception" to this - replicated tables.
  - Data you explicitly want shared across all workers to make for more efficient joins | aggregations | small operations.

Yellowbrick is a Column Store Database

Data Warehouse | Analytic Database | MPP | Shared Nothing | **Column Store** | On-Prem & Cloud

User data is stored in blocks by column, not rows.

What this means:

- You only access the storage for the columns you actually query
  - This makes indexes unnecessary
  - The way you optimized filtering predicates is very different
  - Impacts the creation of good DDLS and ELT
- Blocks contain data of the same type
  - Allows for encoding of data, gaining high data compression.

Yellowbrick is for On-Prem and the cloud

Data Warehouse | Analytic Database | MPP | Shared Nothing | Column Store | **On-Prem & Cloud**

Yellowbrick runs on both hardware in your DC and in the cloud from a common code base.

This allows:

- Customers to run a mixed environment of cloud and on-premises, and even replicate between them.
- The ability to transition to a cloud offering in the future if desired without having to re-architect everything

- Note: Exercises in this training are tailored to the cloud, but the concepts apply to both.

Yellowbrick is …

… a ColumnStore, MPP, Shared Nothing, Analytic, Data Warehouse, for on-prem and cloud

Data Warehouse

Data warehouses often combine data from multiple sources, conform the data to standards, and act as a single source of truth for reporting and analytics.

Shared Nothing

Each Compute Node has autonomy over its subset of data

Analytic Database

A databases designed for analytical queries against large amounts of data.

Column Store

Data storage is in blocks by column not rows.

MPP

Massively Parallel Processing (MPP) databases use an architecture of multiple nodes working together to process queries and data.

On-Prem and in the cloud

Yellowbrick runs on both hardware in your DC and in the cloud from a common code base.

Questions?

Thank You

Your Data Anywhere